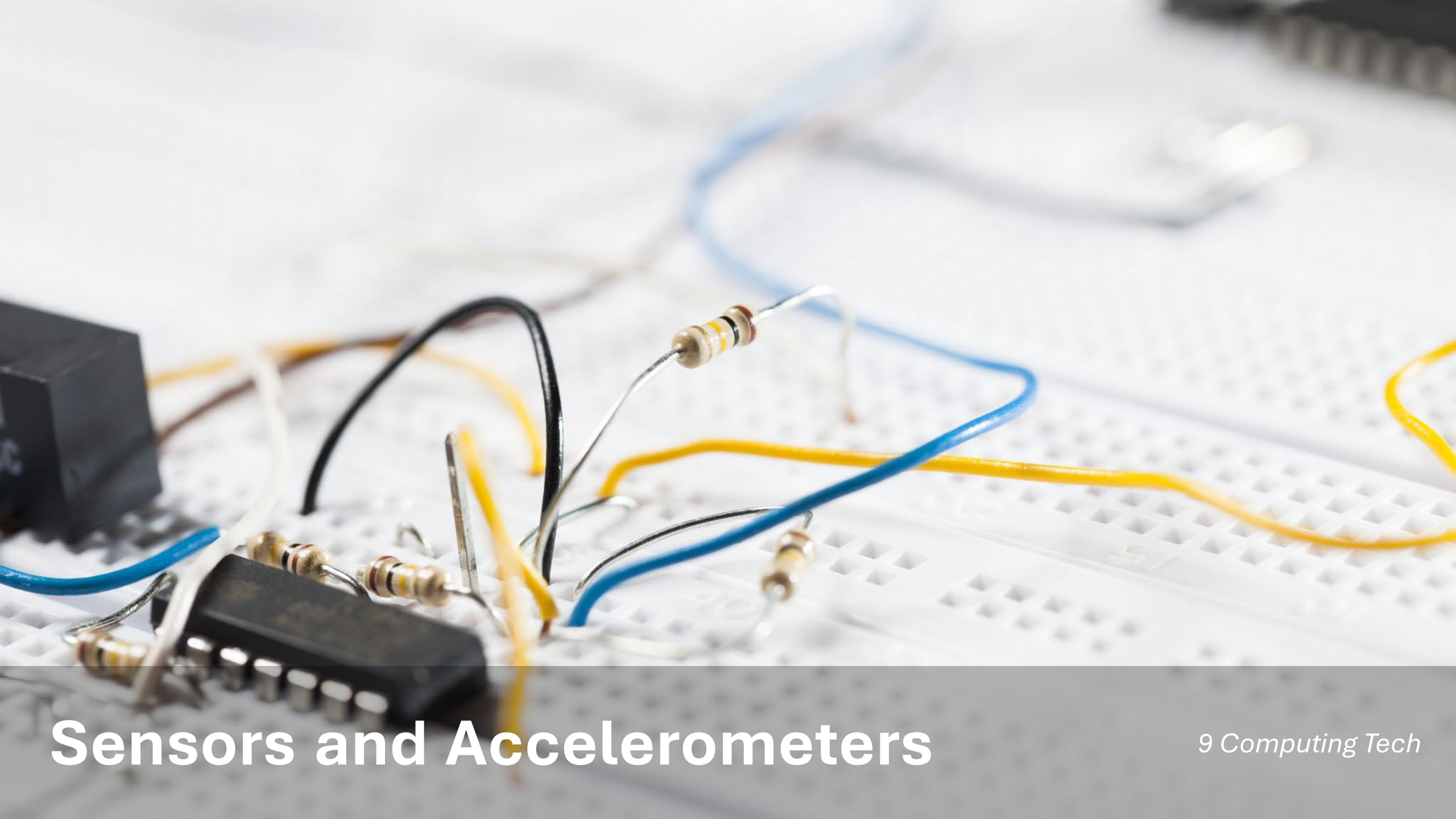


Sensors and Accelerometers

9 Computing Tech

Learning Intention



Describe mechatronic systems



Describe inputs, storage, transmission, processes and outputs in mechatronic and automated systems



Select and justify sensors for a mechatronic or automated system

Success Criteria



I can describe a range of sensor inputs which are commonly used in mechatronic systems.



I can explain the purpose of specific sensor inputs in a mechatronic system.

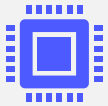
Human Senses vs Sensors



Our senses interpret the world: sight, sound, touch, taste, smell.



Mechatronic sensors are modelled on human senses but have limitations in precision and range.



Sensors are **input devices** that collect data for processing.

Sensors and Mechatronic Systems

Sensor: A device that detects and responds to some type of input from the physical environment.

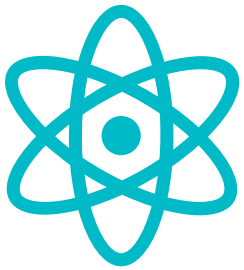


Sensors send data to a microprocessor.

The system processes the data and may trigger an **actuator** or other outputs.

Examples: temperature control, automatic doors, alarms.

Types of Sensors



Analogue vs Digital

Analogue: outputs a continuous range of values (e.g., temperature sensor, light sensor)

Digital: outputs discrete values, usually on/off (e.g., limit switch, push button)



Active vs Passive

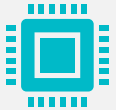
Active: require interaction (e.g., sonar tracking)

Passive: respond to environmental changes (e.g., PIR motion detectors)

Sensor Examples & Functions

Sensor Type	Function	Example Use
Temperature	Measures temperature	Air conditioning control
Motion/Pressure	Detects movement	Security alarms
Optical/Light	Detects light or color changes	Automatic doors
Chemical	Detects chemicals	Hand-held blood tests / air quality

Advanced Sensors – Nanosensors



Tiny, highly sensitive sensors used in medicine and environmental monitoring.



Can detect minute amounts of viruses, bacteria, or toxins.



Example: DNA testing with cantilever nanosensors.



Highlight potential for **nanorobots** in medical applications.

Common Applications



Security Alarms: PIR motion detectors, laser beams, photoelectric sensors



Automatic Doors: Activated by motion or light sensors



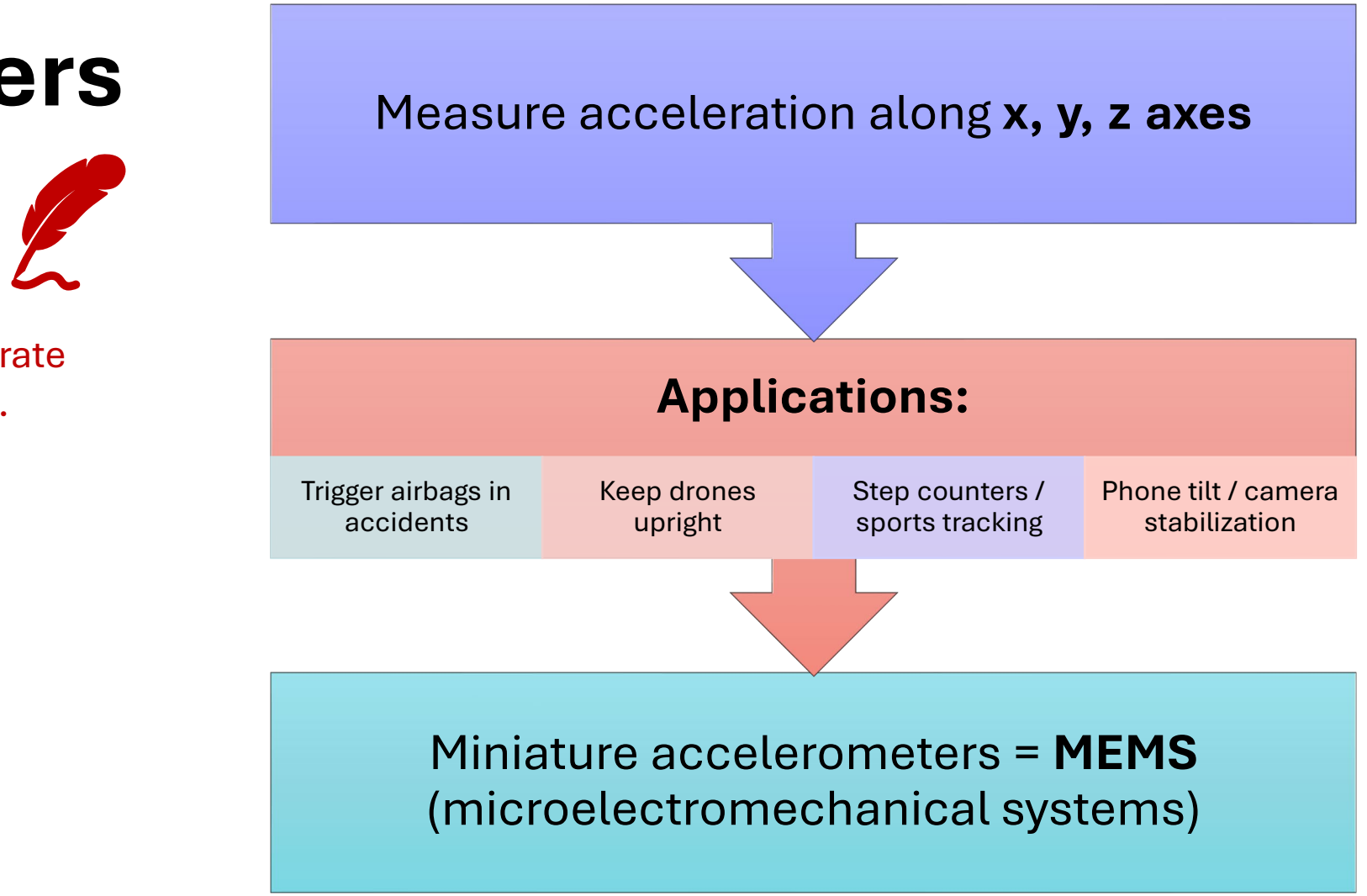
Vehicles: Engine sensors (speed, temperature, fuel, lane detection, airbags)



Phones/Tablets: Accelerometer, gyroscope, magnetometer, proximity sensor

Accelerometers

Accelerometer: a device that measures **acceleration**, which is the rate of change of the **velocity** of an object. They measure in meters per second squared (m/s^2) or in G-forces (g).



Distance Sensors



Infrared (IR): Distance to nearby objects



Ultrasonic: Sound waves measure distance



RADAR: Radio waves measure distance or velocity



LiDAR: Laser light builds 3D maps



Used in production lines, vehicles, and security.

Case Study: Boeing 737 Max

- Tilt sensors gave conflicting readings → MCAS system activated incorrectly
- **Consequences:** repeated nosedives, eventual crash
- **Key lesson:** sensor reliability and error checking is critical in mechatronic systems
- Tie back to importance of proper calibration, redundancy, and verification in sensors.

Data Loggers

Data Logger: a small and relatively inexpensive electronic device that monitors and records data over time

Data Logging: the process of capturing, storing and displaying one or more datasets to analyse activity, trends and predict future events



Electronic devices that use built-in sensors to **record data over time**.



Can store data internally or transmit it to another device (e.g., a PC).



Graphing collected data allows trends to be observed that might otherwise be missed.



Common Uses:

- Weather stations (temperature, rainfall, wind)
- Road traffic monitoring
- Wildlife tracking
- Soil moisture and water levels
- Solar panel electricity production